

## HDOC DAY-15 ACTIVITY

### QUESTION 1 – Menu-Driven Digit Operations

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all digits are even.
  - E.g. 1: Number: 12345, Output: All digits not even
  - E.g. 2: Number: 24680, Output: All digits are even
- Check whether all digits are prime.
  - E.g. 1: Number: 2357, Output: All digits are prime
  - E.g. 2: Number: 2379, Output: All digits are not prime

#### 1. Problem Understanding

- Input: A positive integer and the user's menu choice.
- Output: The result of the operation selected through switch-case.
- For checking prime digits, the prime digits are 2, 3, 5, and 7.

#### Operation Summary

| Choice | Operation                          | Main Logic                                                 |
|--------|------------------------------------|------------------------------------------------------------|
| 1      | Check whether all digits are even  | Extract each digit and check whether it is divisible by 2. |
| 2      | Check whether all digits are prime | Extract each digit and check whether it is 2, 3, 5, or 7.  |
| Other  | Invalid choice                     | Display an appropriate message.                            |

#### 2. Standard Test Cases

| Test Case | Choice | Input     | Expected Output          | Actual Output            | Result |
|-----------|--------|-----------|--------------------------|--------------------------|--------|
| 1         | 1      | n = 24680 | All digits are even      | All digits are even      | Pass   |
| 2         | 1      | n = 12345 | All digits not even      | All digits not even      | Pass   |
| 3         | 2      | n = 2357  | All digits are prime     | All digits are prime     | Pass   |
| 4         | 2      | n = 2379  | All digits are not prime | All digits are not prime | Pass   |

### 3. Algorithm

#### A. Common Menu Logic

1. Start the program.
2. Declare the required variables.
3. Read the positive integer.
4. Display the menu and read the user's choice.
5. Use switch-case to perform the selected operation.
6. Display the result.
7. Stop the program.

#### B. Case 1 – Check Whether All Digits Are Even

8. Initialize an indicator variable all\_even to 1.
9. Extract the last digit using  $n \% 10$ .
10. Check whether the digit is divisible by 2.
11. If the digit is not divisible by 2, set all\_even to 0.
12. Remove the last digit using  $n / 10$  and repeat until all digits are processed.
13. If all\_even is 1, display All digits are even; otherwise display All digits not even.

#### C. Case 2 – Check Whether All Digits Are Prime

14. Initialize an indicator variable all\_prime to 1.
15. Extract the last digit using  $n \% 10$ .
16. Check whether the digit is 2, 3, 5, or 7.
17. If the digit is not one of these prime digits, set all\_prime to 0.
18. Remove the last digit using  $n / 10$  and repeat until all digits are processed.
19. If all\_prime is 1, display All digits are prime; otherwise display All digits are not prime.

### 5. C Program

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
int n, choice, digit, temp;
```

```
int all_even, all_prime;
```

```
printf("Enter a positive integer: ");
```

```
scanf("%d", &n);
```

```
printf("\nMenu\n");
```

```
printf("1. Check whether all digits are even\n");

printf("2. Check whether all digits are prime\n");

printf("Enter your choice: ");

scanf("%d", &choice);

switch (choice)

{

case 1:

temp = n;

all_even = 1;

while (temp > 0)

{

digit = temp % 10;

if (digit % 2 != 0)

all_even = 0;

temp = temp / 10;

}

if (all_even)

printf("All digits are even\n");

else

printf("All digits not even\n");

break;

case 2:

temp = n;

all_prime = 1;

while (temp > 0)

{

digit = temp % 10;
```

```

if (digit != 2 && digit != 3 &&
    digit != 5 && digit != 7)

all_prime = 0;

temp = temp / 10;

}

if (all_prime)

printf("All digits are prime\n");

else

printf("All digits are not prime\n");

break;

default:

printf("Invalid choice.\n");

}

return 0;

}

```

## 6. Compilation and Execution Evidence

```

kali@kali: ~/Desktop
Session Actions Edit View Help

(kali@kali)~/Desktop
$ gedit question.c

** (gedit:2179): WARNING **: 06:27:15.833: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2179): WARNING **: 06:27:15.833: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

(kali@kali)~/Desktop
$ gcc question.c -o question

(kali@kali)~/Desktop
$ ./question
Enter a positive integer: 24680

Menu
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are even

(kali@kali)~/Desktop
$ ./question
Enter a positive integer: 12345

Menu
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits not even

(kali@kali)~/Desktop
$ ./question
Enter a positive integer: 2357

Menu
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are prime

```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
└─$ ./question
Enter a positive integer: 24680

Menu
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are even

(kali@kali)~/Desktop
└─$ ./question
Enter a positive integer: 12345

Menu
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits not even

(kali@kali)~/Desktop
└─$ ./question
Enter a positive integer: 2357

Menu
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are prime

(kali@kali)~/Desktop
└─$ ./question
Enter a positive integer: 2379

Menu
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are not prime

(kali@kali)~/Desktop
└─$
```

## 7. Verification of All Test Cases

| Test Case | Choice | Input     | Expected Output          | Actual Output            | Result |
|-----------|--------|-----------|--------------------------|--------------------------|--------|
| 1         | 1      | n = 24680 | All digits are even      | All digits are even      | Pass   |
| 2         | 1      | n = 12345 | All digits not even      | All digits not even      | Pass   |
| 3         | 2      | n = 2357  | All digits are prime     | All digits are prime     | Pass   |
| 4         | 2      | n = 2379  | All digits are not prime | All digits are not prime | Pass   |